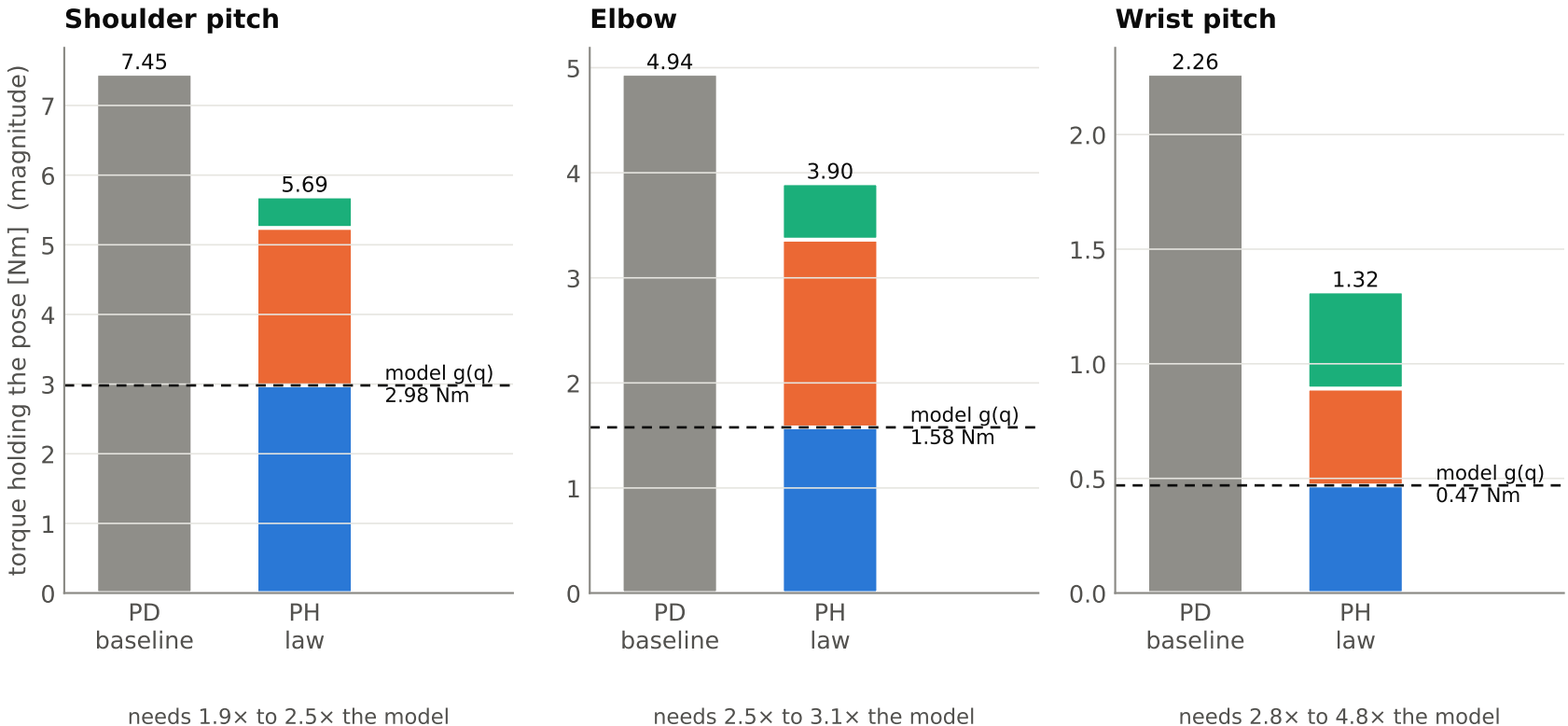
The gravity model under-predicts the torque the arm needs

Torque that actually held the arm still during the hold, from each run. Both runs, measured independently, need 1.9 to 4.8 times Pinocchio's g(q). The larger PH error is what a lower stiffness (18 vs 60 Nm/rad on shoulder pitch) does with the same missing torque.